

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Course Description

Why, What and How — from 2G to 6G

توصیف درس



Lecture Notes

Instructor: Amin Azari

Compiled by Aidin Shekari

استاد: امین آذری github.com/AminAzari



نیمسال پاییز ۱۴۰۵-۱۴۰۶

FALL SEMESTER 2026-2027

In This Module

Introduction	2
Structure and Content	2
Assessment and Practical Work	3
Main References	3

Introduction

Advanced Communication Networks is a 3-credit course (48 hours, 16 weeks $\times$ 3 hours) for senior undergraduate and graduate students of electrical and telecommunication engineering, taught as lectures with laboratories and a project. The course is a coherent technical journey through the evolution of mobile networks from GSM (2G) to LTE (4G), 5G NR/SA and the emerging concepts of 6G — not a collection of separate technologies, but a continuous analysis of **why** each generation appeared, **what** problem it solved and **how** traffic flows through its architecture. The course is built on a “why / what / how” logic so that students can analyse and evaluate architectural decisions instead of memorising lists of protocols.

Structure and Content

The material is organised in **26 modules**:

- **2G/3G foundations (modules 1–4):** GSM architecture, GPRS and EDGE, circuit switching and SS7 signalling, 2G/3G core networks and the basics of mobility management and handover.
- **4G access and core (modules 5–13):** LTE architecture, the E-UTRAN access network, the protocol stack, resource scheduling, the random access channel (RACH), the Evolved Packet Core (EPC), the Diameter protocol, the HSS and authentication, the MME, and the S-GW/P-GW with GTP tunnelling.
- **Software-driven architectures (modules 15–18):** software-defined networking (SDN), network function virtualisation (NFV), the 3GPP standardisation process and the 5G service-based architecture (SBA).
- **5G core and procedures (modules 14, 19–22):** the 5G core architecture, signalling procedures (registration, 5G-AKA authentication, PDU session establishment and release, paging, handover), network slicing and the evolution of the QoS framework (from QCI to 5QI).
- **Planes and outlook (modules 23–26):** the user plane and the control plane, a comprehensive synthesis of network evolution, and 6G research directions including ISAC, RIS, AI-native networks, cell-free MIMO and sub-THz.

Every module comes with comparison tables, architecture diagrams, step-by-step signalling sequence diagrams and review questions. A prerequisites review recalls the basics of OSI, RF, modulation, OFDM, IP networking, probability and decibel arithmetic.

Learning Objectives

By the end of the course, students can trace the evolution of the generations; analyse core architectures (circuit-switched, packet-switched, the all-IP EPC and the service-based 5G core); compare mobility management and handover across generations; apply network slicing, QoS, SDN and NFV in modern networks; explain 5G core procedures at the signalling level; and critically assess 6G research directions.

Assessment and Practical Work

Component	Weight
Exams (midterm 20% + final 30%)	50%
Laboratories (6 labs $\times$ 5%)	30%
Project (proposal 5% + report and demo 15%)	20%

The laboratory assignments cover the LTE link budget, EPC signalling traces, 5G NR scheduling and resource allocation, SBA service discovery, network slicing configuration and user-plane traffic tracing. The project, individual or in pairs, is defined on topics such as a comparative analysis of LTE versus 5G, an enterprise network-slicing design, implementing a scheduling algorithm with the simulator, or signalling analysis with real packet captures.

Main References

Sesia et al. (*LTE – The UMTS Long Term Evolution*), Dahlman et al. (*5G NR: The Next Generation Wireless Access Technology*), Holma and Toskala (*LTE for UMTS*), the 3GPP specifications (TS 23.501, 23.502, 24.501, 38.300) and Rappaport (*Wireless Communications*).